Supplement to "Eco-evolutionary deviations from steady state across the Hawaiian chronosequence from a maximum entropy perspective"

This supplement walks through the steps of recreating the entire analysis presented in the main text. It also provides further analysis in support of and to contextualize results in the main text.

Computing setup

Analyses are supported by a custom R package *HawaiiArthMETE* [@foo] which can be installed from github.

```
devtools::install_github("ecoevomatics/HawaiiArthMETE")
```

The quarto [@quarto] document used to generate this supplement can be found with the R command `system.file("ms/hawaii-arth-mete-supp.qmd", package = "HawaiiArthMETE")`.

The following R computing environment is used for all analyses:

```
library(meteR)
library(ggplot2)
library(ggh4x)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(HawaiiArthMETE)
```

```
Attaching package: 'HawaiiArthMETE'


The following object is masked from 'package:meteR':

    arth
```

```r
knitr::opts_chunk$set(
    # default figure dims
    fig.width = 5.5, fig.height = 4,
    # caching is default for all subsequent chunks
    cache = TRUE
)

# load data
data(arth)

sessionInfo() |>
    print(locale = FALSE)
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.0.1

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

attached base packages:
```

```
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] HawaiiArthMETE_0.1.0 tidyr_1.3.1          dplyr_1.2.1
[4] ggh4x_0.3.1          ggplot2_4.0.1        meteR_1.2

loaded via a namespace (and not attached):
 [1] vctrs_0.7.3        cli_3.6.5          knitr_1.50         rlang_1.2.0
 [5] xfun_0.52          purrr_1.1.0        generics_0.1.4     S7_0.2.1
 [9] jsonlite_2.0.0     glue_1.8.0         htmltools_0.5.8.1  scales_1.4.0
[13] rmarkdown_2.29     grid_4.5.2         evaluate_1.0.4     tibble_3.3.0
[17] fastmap_1.2.0      yaml_2.3.10        lifecycle_1.0.5    compiler_4.5.2
[21] RColorBrewer_1.1-3 pkgconfig_2.0.3    rstudioapi_0.17.1  farver_2.1.2
[25] digest_0.6.39      R6_2.6.1           tidyselect_1.2.1   pillar_1.11.0
[29] magrittr_2.0.4     withr_3.0.2        tools_4.5.2        gtable_0.3.6
```

METE calculations

Comparison of the observed data to predictions from the Maximum Entropy Theory of Ecology (METE) is performed with package *meteR* [@rominger-merow2016]. The following code records information about each sample (name, flow age, proportion non-native presence) as well as several metrics of deviation between theory and observation:

- a z^2 value measuring a normalized difference in likelihoods between observed data and theoretical expectation
- differences in proportion of rarity (number of singletons or normalized metabolic rate ≤ 2) observed and expected
- differences in proportion of dominance (sum of top three most abundant species or sum of top three highest normalized metabolic rates) observed and expected

All metrics are calculated for both the species abundance distribution (SAD) and individual metabolic rate (or power) distribution (IPD).

```r
arth_mete <- split(arth, arth[, c("site", "tree")], drop = TRUE) |>
    lapply(function(x) {
        # the esf for this grouping
        esf <- meteESF(spp = x$species_code,
                       abund = x$abundance,
                       power = x$metabolic_rate)

        # sad and ipd
        s <- sad(esf)
```

```
        p <- ipd(esf)

        data.frame(
            # sample info
            site = x$site[1],
            site_name = x$site_name[1],
            tree = x$tree[1],
            flow_age_my = x$flow_age_my[1],

            # z2 values
            z2_sad = logLikZ(s)$z,
            z2_ipd = logLikZ(p)$z,

            # differences in extreme values
            n1_diff = log(mean(meteDist2Rank(s) == 1) / mean(s$data == 1)),
            nmax_diff = log(sum(meteDist2Rank(s)[1:10]) / sum(s$data[1:10])),
            p1_diff = log(mean(meteDist2Rank(p) <= 3) / mean(p$data <= 3)),
            pmax_diff = log(sum(meteDist2Rank(p)[1:10]) / sum(p$data[1:10]))
        )
    })

arth_mete <- do.call(rbind, arth_mete)
```

Deviations from METE across the chronosequence

Now we analyze how deviations from METE behave across the chronosequence. The following code produces the analysis of a hump-shaped versus flat relationship between METE deviation and flow age.

```
# quadratic model of sad z^2 across age
sad_rma_chrono <- site_means_lm(log(z2_sad) ~
                                    log(flow_age_my) + I(log(flow_age_my)^2) +
                                    (1 | site),
                                data = arth_mete)

# quadratic model of ipd z^2 across age
ipd_rma_chrono <- site_means_lm(log(z2_ipd) ~
                                    log(flow_age_my) + I(log(flow_age_my)^2) +
                                    (1 | site),
                                data = arth_mete)
```

Likelihood ratio tests (LRT) can evaluate the significance of the quadratic model

```r
# likelihood ratio tests for sad and ipd across chrono
sad_lrt_chrono <- anova(sad_rma_chrono)
sad_lrt_chrono
```

```
Test of Moderators (coefficients 2:3):
QM(df = 2) = 26.5750, p-val < .0001
```

```r
ipd_lrt_chrono <- anova(ipd_rma_chrono)
ipd_lrt_chrono
```

```
Test of Moderators (coefficients 2:3):
QM(df = 2) = 3.7035, p-val = 0.1570
```

The quadratic model for SAD z^2-values is supported by the LRT with a P-value < 0.0001, but the quadratic model for IPD z^2-values did not have a significant P-value. However, the 95% CI for the quadratic term in the IPD z^2-values model has the majority of its range below zero, indicating the data do tend toward a quadratic shape, just with greater variance compared to the quadratic model for SAD z^2-values.

```r
ipd_rma_chrono
```

```
Fixed-Effects with Moderators Model (k = 5)

I^2 (residual heterogeneity / unaccounted variability): 0.00%
H^2 (unaccounted variability / sampling variability):   0.08
R^2 (amount of heterogeneity accounted for):            92.14%

Test for Residual Heterogeneity:
QE(df = 2) = 0.1516, p-val = 0.9270

Test of Moderators (coefficients 2:3):
QM(df = 2) = 3.7035, p-val = 0.1570

Model Results:
```

```
                        estimate      se     zval    pval    ci.lb   ci.ub
intrcpt                  -0.0665  0.4239  -0.1568  0.8754  -0.8972  0.7643
log(flow_age_my)         -0.4871  0.2561  -1.9020  0.0572  -0.9890  0.0148  .
I(log(flow_age_my)^2)    -0.0645  0.0342  -1.8879  0.0590  -0.1314  0.0025  .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The following code produces the visualization reported in the main text of deviation from METE across flow ages.

```
# prepare data frame of predicted values for plotting
x <- range(arth$flow_age_my) |>
    log() |>
    (\(x) seq(x[1], x[2], length.out = 50))() |>
    exp() |>
    data.frame(flow_age_my = _)

sad_hat <- site_means_predict(sad_rma_chrono, newdata = x)
ipd_hat <- site_means_predict(ipd_rma_chrono, newdata = x)

z2_chrono_pred <- bind_rows(z2_sad = data.frame(x, sad_hat),
                            z2_ipd = data.frame(x, ipd_hat),
                            .id = "metric") |>
    select(!se) |>
    rename(z2 = pred,
           z2_lo = ci.lb,
           z2_hi = ci.ub) |>
    mutate(across(starts_with("z2"), exp))


# plotting
fig_z2_chrono <- pivot_longer(arth_mete, c(z2_sad, z2_ipd),
             names_to = "metric", values_to = "z2") |>
    ggplot(aes(flow_age_my, z2, color = site_name)) +
    geom_jitter(width = 0.1) +
    # confidence envelope
    geom_ribbon(data = z2_chrono_pred,
                mapping = aes(x = flow_age_my,
                              ymin = z2_lo,
                              ymax = z2_hi),
                color = "transparent", alpha = 0.4) +
```

```r
    # central tendency
    geom_line(data = z2_chrono_pred,
              mapping = aes(flow_age_my, z2),
              color = "black") +
    # facet by IPD or SAD and trick strip label into axis label
    facet_grid(
        rows = vars(metric),
        labeller = as_labeller(
            c(z2_ipd = "\"IPD\"~z^2*\"-value\"",
              z2_sad = "\"SAD\"~z^2*\"-value\""),
            label_parsed
        ),
        switch = "y"
    ) +
    xlab("Flow age (my)") +
    this_ms_look(log_x = TRUE, log_y = TRUE) +
    theme(axis.title.y = element_blank(),
          strip.background = element_blank(),
          strip.placement = "outside")

fig_z2_chrono
```

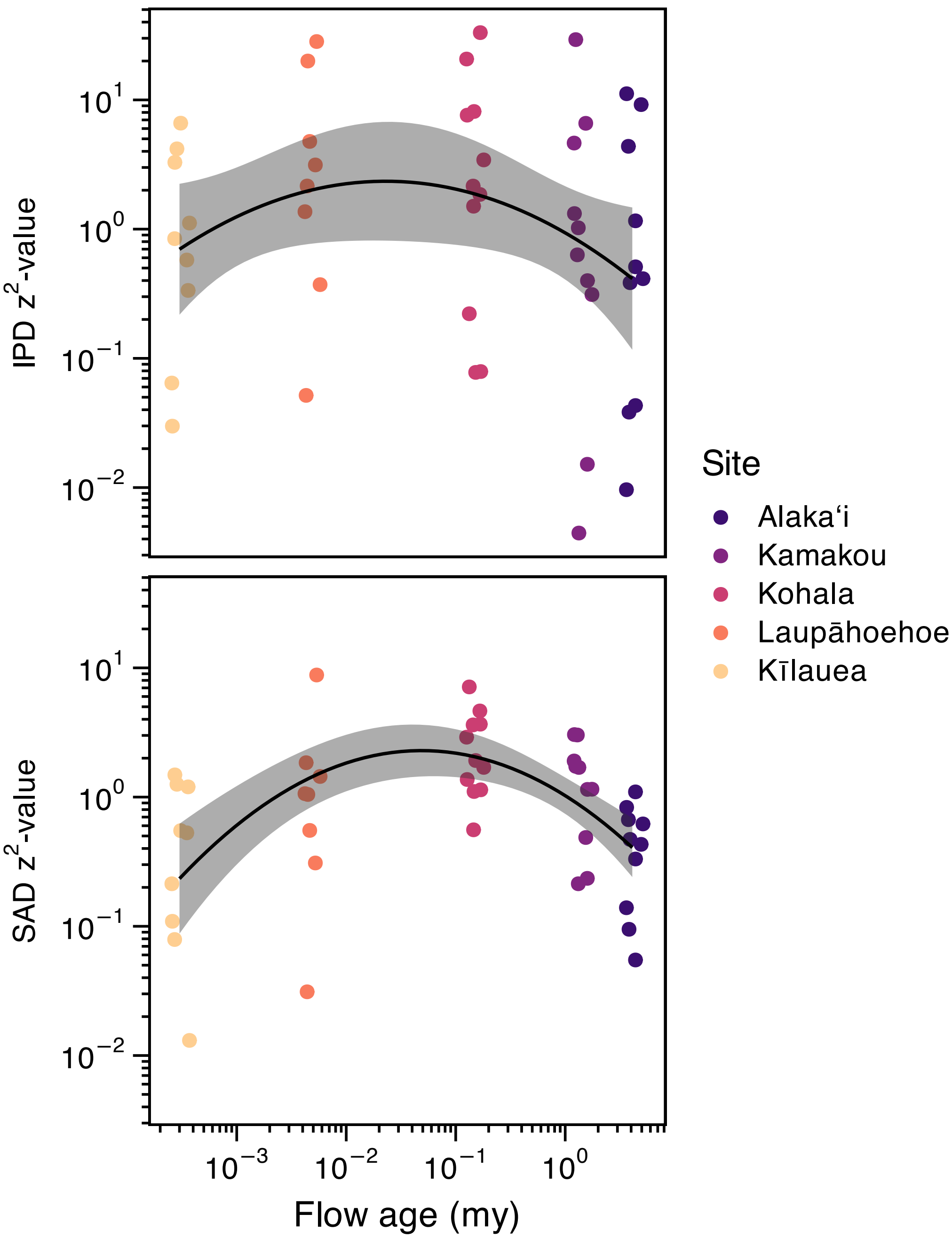

Figure 1: Deviations from METE as measured by z^2 across the chronosequence. Discussed in more detail in the main text.

How do observed patterns deviate from METE?

Next we determine how the observed data deviate from the METE predictions. To begin we look at two exemplar SADs and two exemplar IPDs representing low and high deviation from theory (Figure 2).

```
# function to summarize theoretical and observed distributions
# given one sample `x`

mk_summ <- function(x) {
    esf <- meteESF(x$species_code, x$abundance, x$metabolic_rate)

    list(sad = sad(esf),
         ipd = ipd(esf)) |>
        lapply(function(x) {
            thr <- samp_mete(x, n = 1000) |>
                summarize_samp_mete()

            obs <- data.frame(rank = seq_along(x$data),
                              abund_obs = x$data)

            full_join(thr, obs)
        }) |>
        bind_rows(.id = "dist")
}



expl_samps <- data.frame(site = c("KA", "MO"),
                         tree = c(5, 9),
                         deviation = c("lo", "hi")) |>
    left_join(arth) |>
    (\(x) split(x, x$deviation, drop = TRUE))() |>
    lapply(mk_summ) |>
    bind_rows(.id = "deviation")
```

```
Joining with `by = join_by(site, tree)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
```

```r
fig_how_diff <- ggplot(expl_samps, aes(rank, m)) +
    geom_point(aes(rank, abund_obs)) +
    geom_ribbon(aes(ymin = lo, ymax = hi),
                fill = "#489BC5", alpha = 0.75) +
    geom_line(color = "#489BC5") +
    facet_grid2(
        rows = vars(dist), cols = vars(deviation),
        scales = "free", independent = "x",
        labeller = as_labeller(
            c(hi = "High deviation", lo = "Low deviation",
              ipd = "Metabolic rate", sad = "Abundance")
        ),
        switch = "y"
    ) +
    xlab("Rank") +
    ylab("") +
    this_ms_look(log_y = TRUE) +
    theme(strip.background.y = element_blank(),
          strip.placement.y = "outside",
          strip.text.y = element_text(size = 14))

fig_how_diff
```

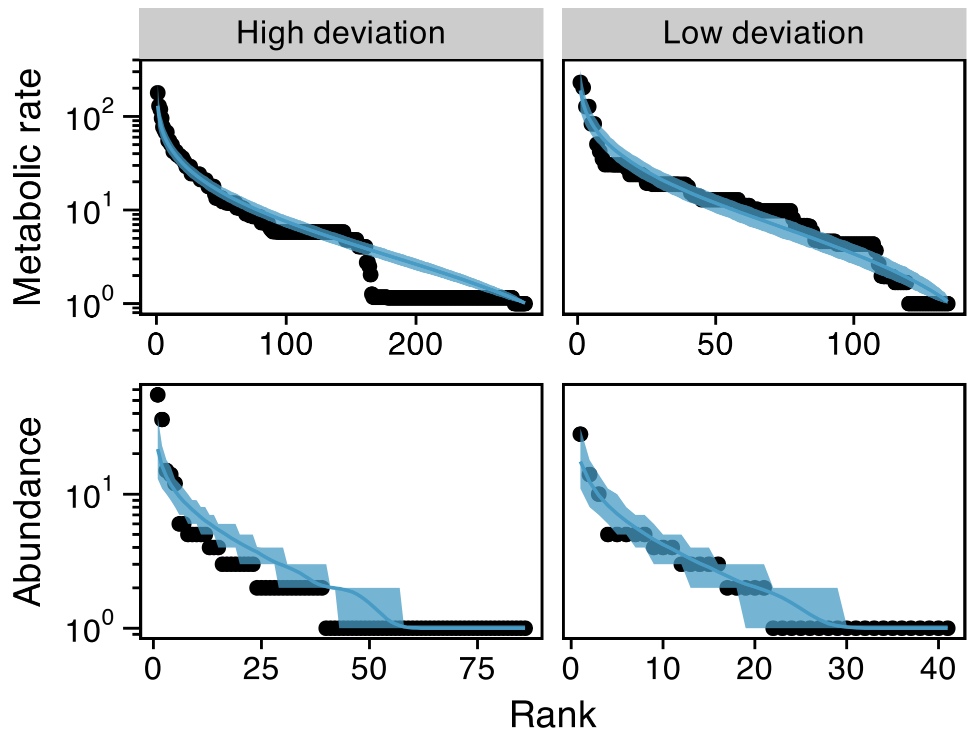

Figure 2: Two exemplar IPDs and SADs showing high and low deviation from METE. The low deviation distributions come from the same sample at the Alakaʻi site while the high deviation distributions come from the same sample at the Kamakou site. Samples were choosen to represent ranges of deviation while still being similar in state variables

Looking across all samples (Figure 3) we see that deviations between observed and theoretical SADs is consistently in the direction of under-predicted both rarity and dominance (i.e. observed SADs have more rarity and greater dominance than theoretical predictions). Indeed, only one sample shows a (small) over-prediction in dominance, and only two samples show a (small) over-prediction in rarity. This means that observed SADs are more uneven than the METE prediction.

```
fig_how_z2 <- bind_rows(sad = select(arth_mete, site_name,
                        z2 = z2_sad, rar_diff = n1_diff, dom_diff = nmax_diff),
          ipd = select(arth_mete, site_name,
                        z2 = z2_ipd, rar_diff = p1_diff, dom_diff = pmax_diff),
          .id = "dist") |>
    pivot_longer(ends_with("diff"),
                 names_to = c("comp", ".value"),
                 names_sep = "_") |>
```

```r
    ggplot(aes(diff, z2, color = site_name)) +
    geom_point() +
    facet_grid2(rows = vars(dist), cols = vars(comp),
                scales = "free", independent = "x",
                labeller = as_labeller(
                    c(ipd = "\"IPD\"~z^2*\"-value\"",
                      sad = "\"SAD\"~z^2*\"-value\"",
                      dom = "\"Diff. in dominance\"",
                      rar = "\"Diff. in rarity\""),
                    label_parsed
                ),
                switch = "both") +
    xlab("") +
    ylab("") +
    this_ms_look(log_y = TRUE) +
    geom_vline(xintercept = 0, color = "gray") +
    theme(strip.background = element_blank(),
          strip.placement = "outside",
          strip.text.y = element_text(size = 14))


fig_how_z2
```

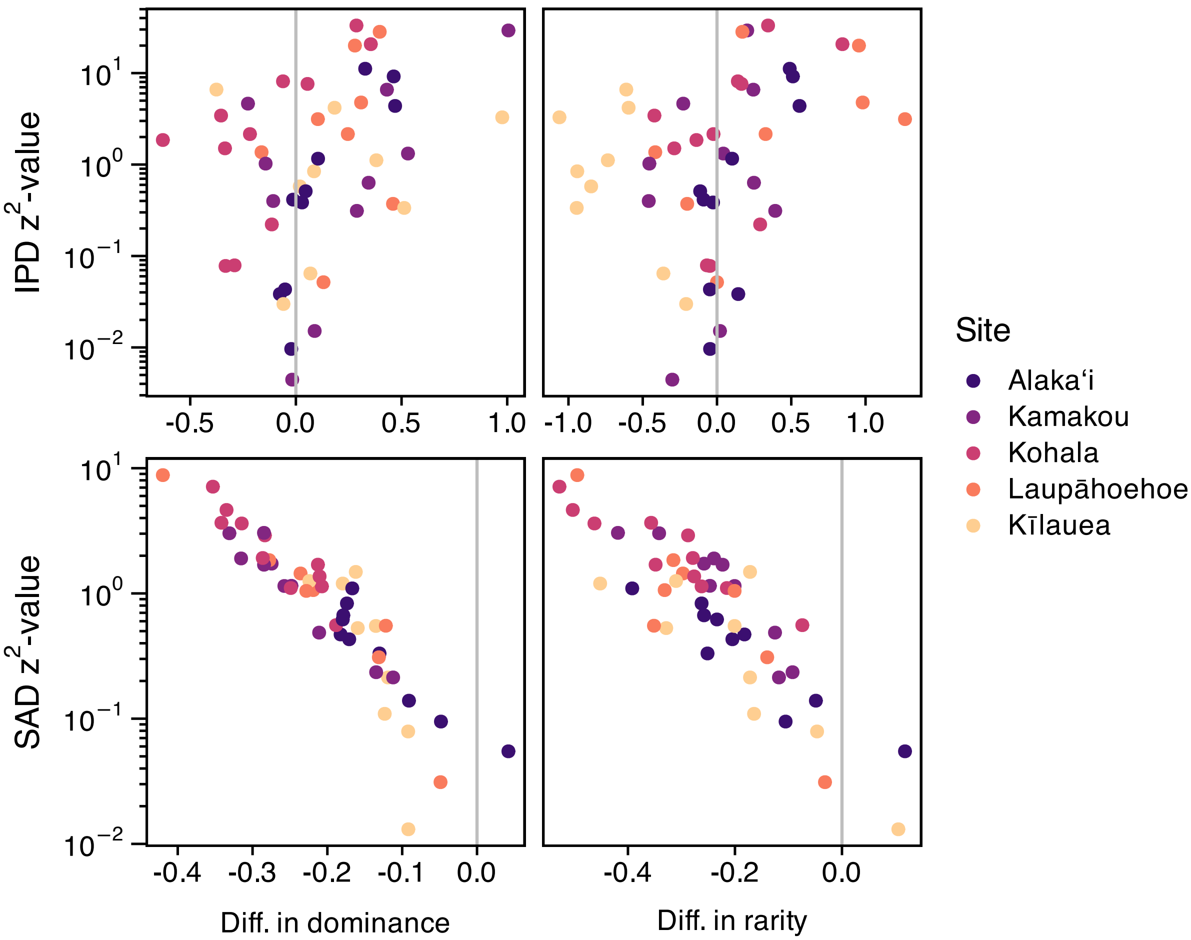

Figure 3: Relationship between z^2 measure of deviation from METE and different metrics describing how observed data and METE predictions diverge.

In the case of the IPD dominance refers to metabolic rates of individuals with the highest metabolic rates, and rarity refers to the number of individuals with very low metabolic rates. Actual and theoretical IPDs do not show this consistent discrepancy between observed and predicted dominance and rarity. To evaluate this we model the probability of whether the log ratio is above or below zero with an intercept only binomial GLM.

```
# "successes" and "failures" for GLM
ipd_diff <- summarize(arth_mete,
                      n_over_dom = sum(pmax_diff < 0),
                      n_undr_dom = sum(pmax_diff > 0),
                      n_over_rar = sum(p1_diff < 0),
                      n_undr_rar = sum(p1_diff > 0),
                      .by = site)
```

```r
# dominance
pmax_mod <- glm(cbind(n_over_dom, n_undr_dom) ~ 1,
                family = binomial, data = ipd_diff)

summary(pmax_mod)
```

```
Call:
glm(formula = cbind(n_over_dom, n_undr_dom) ~ 1, family = binomial,
    data = ipd_diff)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.4229     0.2951  -1.433    0.152

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 9.069  on 4  degrees of freedom
Residual deviance: 9.069  on 4  degrees of freedom
AIC: 23.517

Number of Fisher Scoring iterations: 3
```

```r
# rarity
p1_mod <- glm(cbind(n_over_rar, n_undr_rar) ~ 1,
              family = binomial, data = ipd_diff)

summary(p1_mod)
```

```
Call:
glm(formula = cbind(n_over_rar, n_undr_rar) ~ 1, family = binomial,
    data = ipd_diff)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.2136     0.2934   0.728    0.467

(Dispersion parameter for binomial family taken to be 1)
```

```
    Null deviance: 13.766  on 4  degrees of freedom
Residual deviance: 13.766  on 4  degrees of freedom
AIC: 26.51

Number of Fisher Scoring iterations: 4
```

The fact that in both models the intercept is not significantly different from the null ($p = 0.5$) means that there is not evidence of systematic difference between thertical prediction and observed IPD.

Comparing deviations from METE with potential explanatory variables

Deviation vs. proportion of non-native species

To see if proportion of non-native species has any relationship with deviation from METE we use meta regression as with the chronosequence analysis. The below code produces this meta regression and tests the affect of proportion of non-native species on deviations of the SAD and IPD from METE predictions.

```r
# calculate proportion of non-native spp per site
mete_by_non <- summarize(
    arth,
    prop_nspp_non_nat = n_distinct(
        species_code[!(origin %in% c("endemic", "indigneous"))]) /
        n_distinct(species_code),
    .by = site_name) |>
    # join to mete summary statistics
    full_join(arth_mete)
```

```
Joining with `by = join_by(site_name)`
```

```r
# sad regression
sad_rma_non <- site_means_lm(log(z2_sad) ~ prop_nspp_non_nat + (1 | site),
                             # excluding Kīlauea site because it is
                             # quite different, further assessed below
                             filter(mete_by_non, site != "VO"))

# sad likelihood ratio test
anova(sad_rma_non)
```

```
Test of Moderators (coefficient 2):
QM(df = 1) = 19.7465, p-val < .0001
```

```r
# ipd regression
ipd_rma_non <- site_means_lm(log(z2_ipd) ~ prop_nspp_non_nat + (1 | site),
                             # excluding Kīlauea site
                             filter(mete_by_non, site != "VO"))

# ipd likelihood ratio test
anova(ipd_rma_non)
```

```
Test of Moderators (coefficient 2):
QM(df = 1) = 2.7358, p-val = 0.0981
```

The below code is used for visualizing the results. Note that Kīlauea is considered separately because it falls clearly below the expected trend based on other sites. In the case of the SAD, the mean and 95% CI of the z^2-value for Kīlauea is below both the expected value and 95% CI of the rest of the data. In the case of the Kīlauea IPD only the mean of z^2-values is below the 95% CI of the rest of the data; the two CIs do overlap. This is why in the meta regression analysis Kīlauea was excluded, so we could interrogate the way it deviates from the overall pattern.

```r
# 95% CI for mean of Kīlauea site
vo_mean <- filter(mete_by_non, site == "VO") |>
    pivot_longer(starts_with("z2"),
                 names_to = c(".value", "dist"), names_sep = "_") |>
    mutate(z2 = log(z2)) |>
    summarize(site_name = first(site_name),
              prop_nspp_non_nat = mean(prop_nspp_non_nat),
              z2_mean = mean(z2),
              z2_se = sd(z2) / sqrt(n()),
              z2_lo = z2_mean - 1.96 * z2_se,
              z2_hi = z2_mean + 1.96 * z2_se,
              .by = dist) |>
    rename(z2 = z2_mean) |>
    mutate(across(starts_with("z2"), exp))

# prediction for plotting
x <- range(mete_by_non$prop_nspp_non_nat) |>
```

```
    (\(x) seq(x[1], x[2], length.out = 50))() |>
    data.frame(prop_nspp_non_nat = _)

sad_hat <- site_means_predict(sad_rma_non, newdata = x)
ipd_hat <- site_means_predict(ipd_rma_non, newdata = x)

z2_non_pred <- bind_rows(sad = data.frame(x, sad_hat),
                         ipd = data.frame(x, ipd_hat),
                         .id = "dist") |>
    select(!se) |>
    rename(z2 = pred,
           z2_lo = ci.lb,
           z2_hi = ci.ub) |>
    mutate(across(starts_with("z2"), exp))

# plotting
fig_dev_non <- pivot_longer(mete_by_non, starts_with("z2"),
                            names_to = c(".value", "dist"),
                            names_sep = "_") |>
    ggplot(aes(prop_nspp_non_nat, z2, color = site_name)) +
    geom_jitter(width = 0.002) +
    # confidence envelope
    geom_ribbon(data = z2_non_pred,
                mapping = aes(x = prop_nspp_non_nat,
                              ymin = z2_lo,
                              ymax = z2_hi),
                color = "transparent", alpha = 0.4) +
    # central tendency
    geom_line(data = z2_non_pred,
              mapping = aes(prop_nspp_non_nat, z2),
              color = "black") +
    geom_rect(data = vo_mean,
              mapping = aes(xmin = prop_nspp_non_nat - 0.002,
                            xmax = prop_nspp_non_nat + 0.002,
                            ymin = z2_lo,
                            ymax = z2_hi),
              color = "transparent", alpha = 0.4) +
    geom_segment(data = vo_mean,
              mapping = aes(x = prop_nspp_non_nat - 0.002,
                            xend = prop_nspp_non_nat + 0.002,
                            y = z2,
                            yend = z2),
```

```
              color = "black") +
    facet_grid(rows = vars(dist),
               labeller = as_labeller(
                   c(ipd = "\"IPD\"~z^2*\"-value\"",
                     sad = "\"SAD\"~z^2*\"-value\""),
                   label_parsed
               ),
               switch = "y") +
    xlab("Proportion of non-native spp.") +
    this_ms_look(log_y = TRUE) +
    theme(axis.title.y = element_blank(),
          strip.background = element_blank(),
          strip.placement = "outside")



fig_dev_non
```

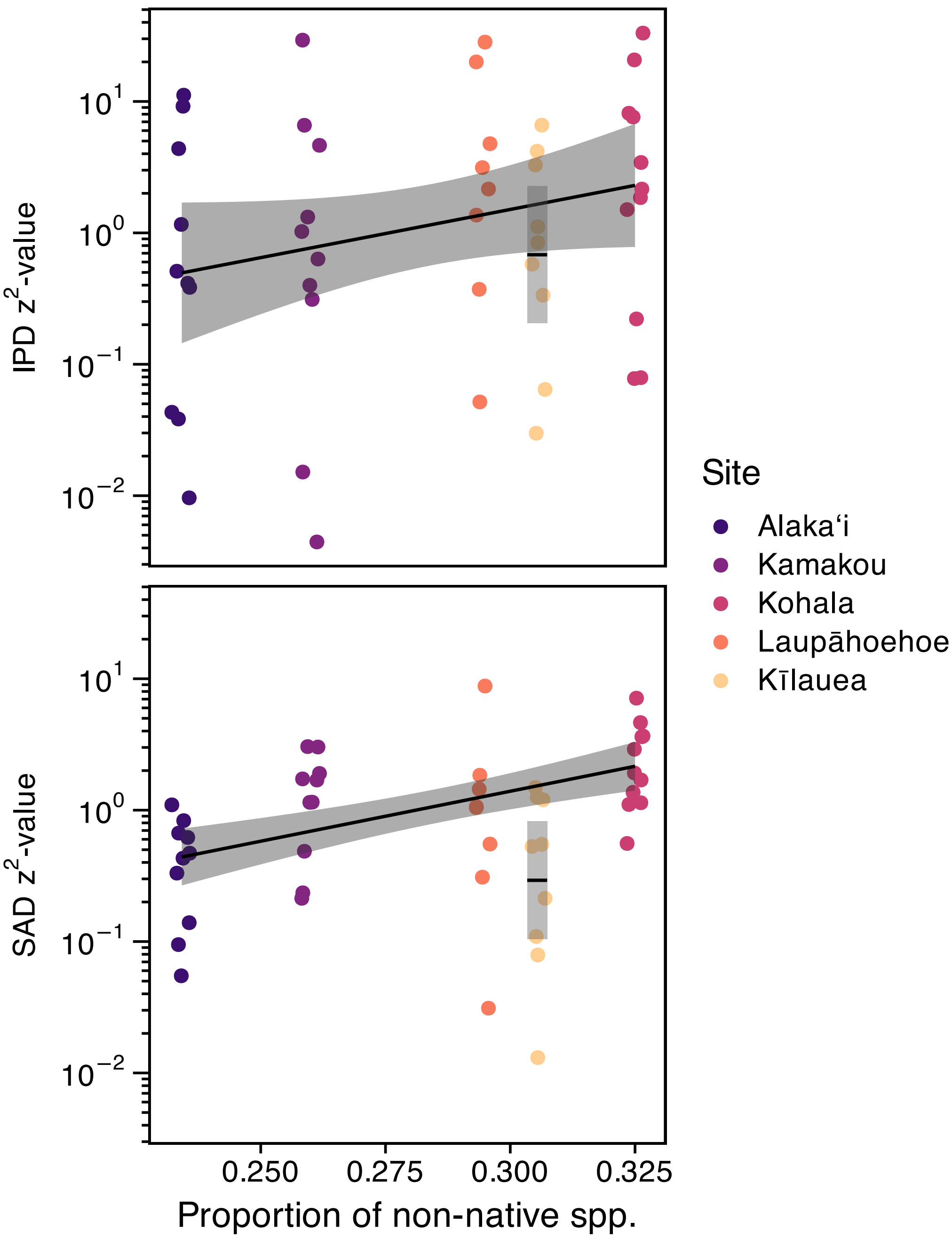

Figure 4: Relationship between proportion of non-native species and deviation from METE. Kīlauea was excluded from the overall analysis to be able to compare it separately to the overall trend. Gray regreions represent 95% confidence envelopes (or confidence interval for Kīlauea) and solid lines are expected values.

Deviation vs. dissimilarity

We also conducted meta regression to see if z^2-values change meaningfully across different levels of within-site, between-sample dissimilarity or β-diversity. The below code runs this meta regression analysis.

```
# calculate dissimilarity z-scores
diss <- split(arth, arth$site) |>
    sapply(FUN = function(dat) {
        summarize(dat, abund = n(), .by = c(tree, species_code)) |>
            pivot_wider(names_from = species_code,
                        values_from = abund,
                        values_fill = 0) |>
            select(!tree) |>
            bray_diss_z()
    })

# combine with mete summary statistics
mete_by_diss <- mutate(arth_mete,
                       mean_diss = diss[site])



# sad regression
sad_rma_diss <- site_means_lm(log(z2_sad) ~ mean_diss + (1 | site),
                              data = mete_by_diss)

# sad likelihood ratio test
anova(sad_rma_diss)
```

```
Test of Moderators (coefficient 2):
QM(df = 1) = 27.7000, p-val < .0001
```

```
# ipd regression
ipd_rma_diss <- site_means_lm(log(z2_ipd) ~ mean_diss + (1 | site),
                              data = mete_by_diss)

# ipd likelihood ratio test
anova(ipd_rma_diss)
```

```
Test of Moderators (coefficient 2):
QM(df = 1) = 2.6460, p-val = 0.1038
```

The relationship between deviation from METE and dissimilarity is more clearly linear than that of proportion non-native species and deviation from METE. It is therefore worth figuring out if the ways that native and non-native taxa contribute to dissimilarity is different across sites.

The below code visualizes the results of the meta regression or SAD and IPD z^2-values against dissimilarity.

```
# prediction for plotting
x <- range(mete_by_diss$mean_diss) |>
    (\(x) seq(x[1], x[2], length.out = 50))() |>
    data.frame(mean_diss = _)

sad_hat <- site_means_predict(sad_rma_diss, newdata = x)
ipd_hat <- site_means_predict(ipd_rma_diss, newdata = x)

z2_diss_pred <- bind_rows(sad = data.frame(x, sad_hat),
                          ipd = data.frame(x, ipd_hat),
                          .id = "dist") |>
    select(!se) |>
    rename(z2 = pred,
           z2_lo = ci.lb,
           z2_hi = ci.ub) |>
    mutate(across(starts_with("z2"), exp))

# plotting
fig_diss_z <- pivot_longer(mete_by_diss, starts_with("z2"),
                           names_to = c(".value", "dist"), names_sep = "_") |>
    ggplot(aes(mean_diss, z2, color = site_name)) +
    geom_jitter(width = 0.1) +
    # confidence envelope
    geom_ribbon(data = z2_diss_pred,
                mapping = aes(x = mean_diss,
                              ymin = z2_lo,
                              ymax = z2_hi),
                color = "transparent", alpha = 0.4) +
    # central tendency
    geom_line(data = z2_diss_pred,
              mapping = aes(mean_diss, z2),
              color = "black") +
```

```
    facet_grid(rows = vars(dist),
               labeller = as_labeller(
                   c(ipd = "\"IPD\"~z^2*\"-value\"",
                     sad = "\"SAD\"~z^2*\"-value\""),
                   label_parsed
               ),
               switch = "y") +
    xlab("Mean dissimilarity") +
    this_ms_look(log_y = TRUE) +
    theme(axis.title.y = element_blank(),
          strip.background = element_blank(),
          strip.placement = "outside")

fig_diss_z
```

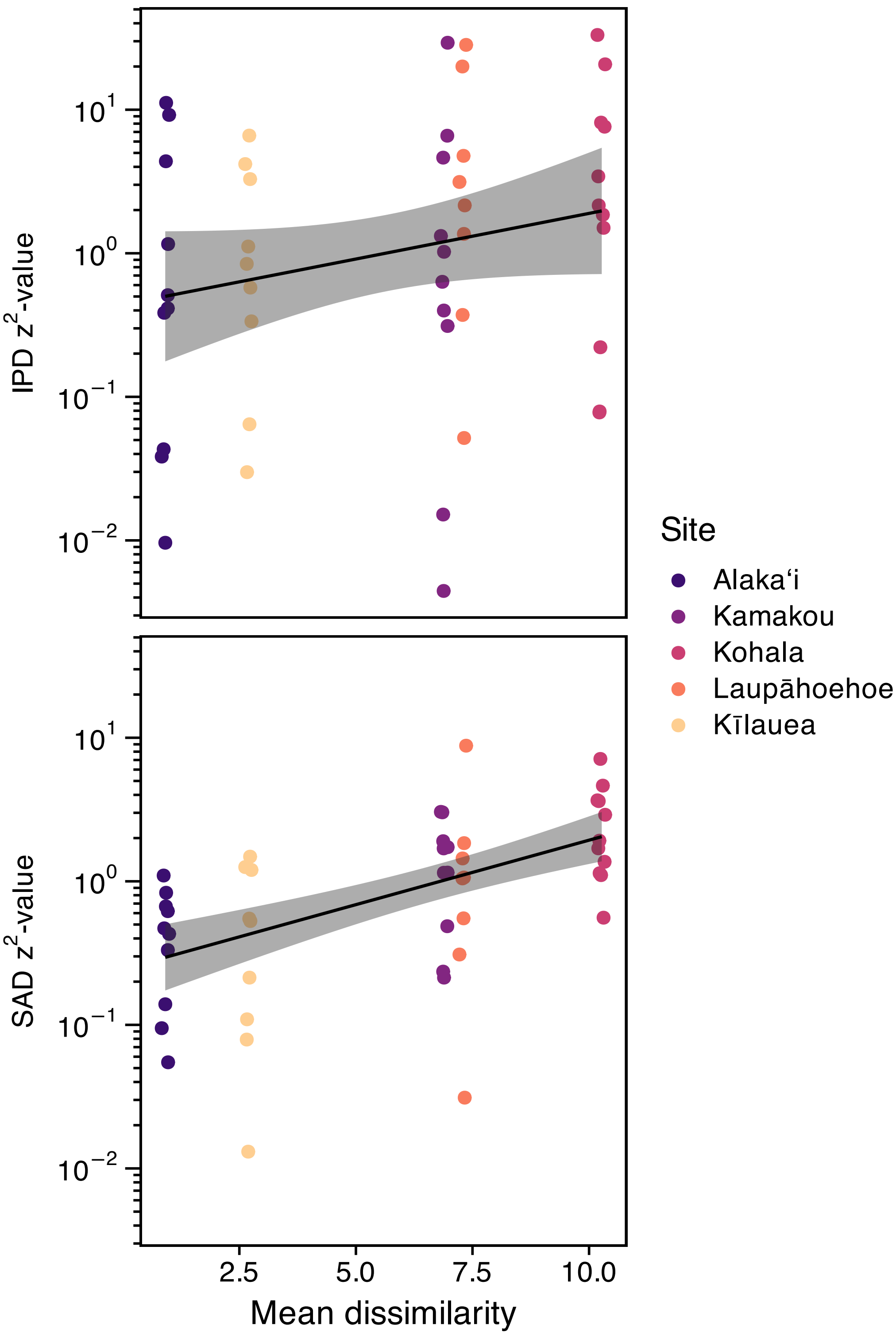

Figure 5: Relationship between dissimilarity and deviation from METE. Gray regreions represent 95% confidence envelopes and solid lines are expected values.

Partitioning Bray Curtis dissimilarity between native and non-native species

To do that we take advantage of the additive nature of Bray Curtis dissimilarity. Bray Curtis dissimilarity is defined as

$$d_{kj} = \frac{\sum_i |x_{ij} - x_{ik}|}{\sum_{ij} x_{ij} + x_{ik}}$$

where x_{ij} and x_{ik} are the abundances of species i at sites j and k. This can be partitioned across any grouping of species, say (a) and (b) with abundances $x_{ij}^{(a)}$ and $x_{ij}^{(b)}$

$$d_{kj} = \frac{\sum_i |x_{ij}^{(a)} - x_{ik}^{(a)}|}{\sum_{ij} x_{ij} + x_{ik}} + \frac{\sum_i |x_{ij}^{(b)} - x_{ik}^{(b)}|}{\sum_{ij} x_{ij} + x_{ik}}$$

Now we can look separately at the contribution of group (a) and group (b) to the overall dissimilarity. We use this approach to compare the contributions of native and non-native taxa to overall dissimilarity. We again make use of a permutational null model to compare these contributions. This null model cannot be the same as the null model used to calculate z scores for overall dissimilarity because that null model shuffled species across samples and individuals across species. Instead, we want to shuffle group labels (native or non-native) across species. The custom function `bray_part_z` calculates these z scores using another custom function `cswap` from *HawaiiArthMETE*.

```r
site_bray_z <- split(arth, arth$site_name) |>
    lapply(function(dat) {
        comm <- summarize(dat,
                    abund = n(),
                    .by = c(tree, species_code)) |>
            pivot_wider(names_from = species_code,
                        values_from = abund,
                        values_fill = 0) |>
            select(!tree)
        g <- dat$origin[match(names(comm), dat$species_code)] %in%
            c("endemic", "indigenous")

        data.frame(bp = bray_part_z(as.matrix(comm), g))
    }) |>
    bind_rows(.id = "site_name")
```

Now we can visualize different contributions of native and non-native species to within-site, between-sample dissimilarity.

```r
site_bray_z$site_name <- factor(site_bray_z$site_name,
                                levels = levels(arth$site_name))

fig_bray_part <- ggplot(site_bray_z, aes(bp, site_name, fill = site_name)) +
    geom_bar(stat = "identity") +
    this_ms_look() +
    geom_vline(xintercept = 0, color = "gray") +
    xlim(c(-1, 1) * max(abs(site_bray_z$bp))) +
    ylab("") +
    xlab("Log ratio dissimilarity partition") +
    annotate("segment", x = c(-0.5, 0.5), xend = c(-2.75, 2.75),
             y = 6, yend = 6,
             arrow = arrow(length = unit(1, "char"))) +
    annotate("text", x = c(-3, 3), y = 6,
             label = c("More non-native\ncontribution",
                       "More native\ncontribution"),
             hjust = c(1, 0), size = 4) +
    theme(legend.position = "none",
          plot.margin = margin(40, 40, 7, 7)) +
    coord_cartesian(ylim = c(1, 5), clip = "off")

fig_bray_part
```

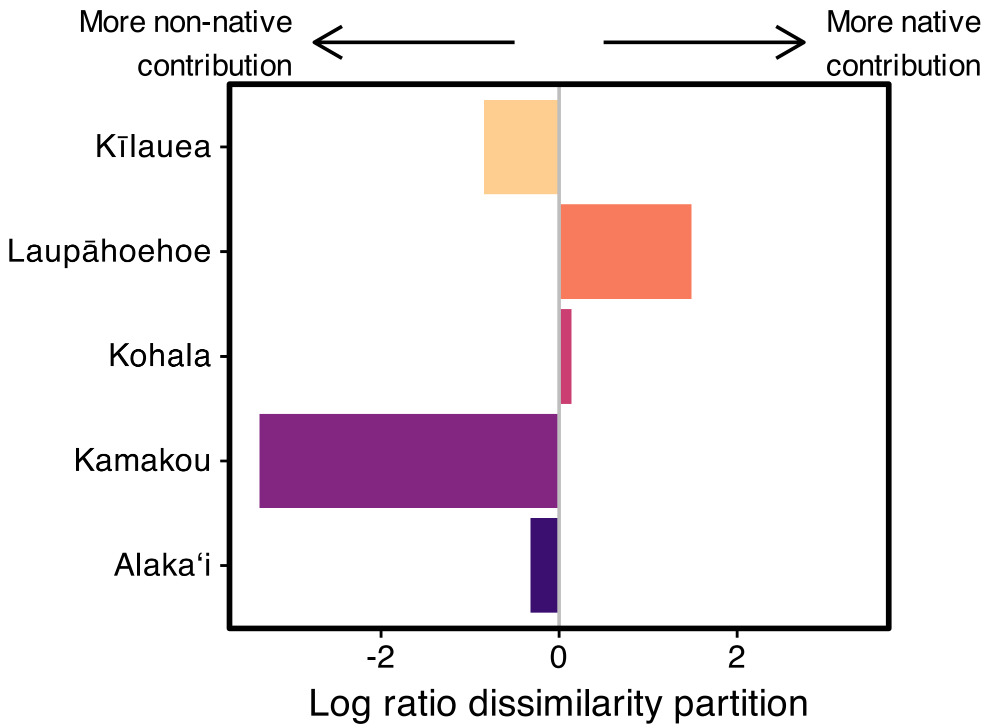

Figure 6: Difference in contribution of native and non-native species to dissimilarity compared to null. Negative values indicate greater contribution of non-native species compared to chance while positive values indicate greater contribution of native species compared to chance.